Predicting Molecular Photostability (T80)

One-page summary - Kaggle molecular-machine-learning - 42 labelled molecules, 9 to predict, metric MSLE (lower is better)

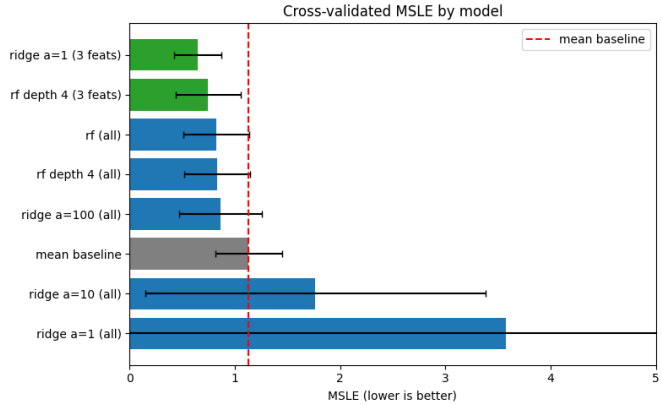
42 / 143	0.647	-43%	0.680
training molecules / features	best CV MSLE (Ridge, 3 features)	error vs. mean baseline (1.133)	leave-one-out MSLE of final blend

Approach

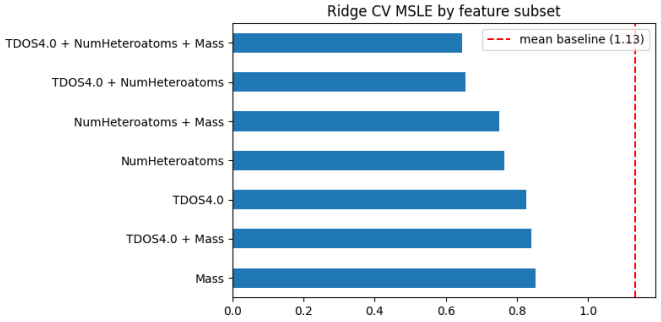
Train on log(1 + T80) so that mean squared error equals the competition metric. Compare models with repeated 5-fold cross-validation (15 scores each), never with training error. Screen all 143 features with random forest importance, permutation importance and Lasso bootstrap stability, then validate small feature sets.

Key findings

- **Fewer features win.** Ridge on TDOS4.0 + NumHeteroatoms + Mass scores 0.647; the best all-feature model (random forest) scores 0.821, and weakly regularised all-feature ridge (3.58) is three times worse than guessing the mean.
- **Two features carry the signal:** TDOS4.0 and NumHeteroatoms, both negatively related to T80 (average importance ranks 3.0 and 4.0 of 143). Together 0.655; Mass adds only 0.008.
- **Chemistry:** the most stable molecules are alkylated thiophenes with only two heteroatoms (T80 ~101-104); molecules with sulfonamide, carbonyl or carbamate groups are less stable.
- **Mass is tricky:** negative on its own, positive in the model (+0.277) because it correlates with heteroatom count. It does not mean heavier is more stable.
- **Not helpful:** symbolic regression (1.252, worse than baseline); automatic feature selection in CV (0.93-1.09).



Cross-validated MSLE by model (green = 3 features)



Ridge CV MSLE by subset of known features

Submission (70% Ridge + 30% random forest, 3 features) - predicted T80

Test-01	02	03	04	05	06	07	08	09
35.9	**30.9**	**19.8**	**26.4**	**38.0**	**36.4**	**31.1**	**20.2**	**22.7**

Predictions span 19.8 to 38.0. They are compressed toward the centre and sit above the training mean (22.7) because the test molecules are small and simple. Test-01 and Test-07 are the least certain: Ridge and forest disagree by about 2x.

Caveats

Expect a typical error of about a factor of 2 (RMSLE ~0.8). Only 42 training and 9 test molecules, so scores are noisy. The blend beats Ridge alone by just 0.005 (0.680 vs 0.685). The best brute-force subset (0.603) is optimistically biased and was not used. Findings are associations, not mechanisms.